

TAN JUIN-E SAMUEL

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EDUCATION

Nanyang Technological University

Bachelor of Computing in Computer Science, Second Major in Business

Aug 2024 - Jul 2028

- Grade Point Average: 4.51 / 5.00
- Recipient, NTU School Scholarship
- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, C/C++ Programming, Operating Systems, Computer Networks, Software Engineering, Probability & Statistics, Discrete Mathematics

WORK EXPERIENCE

Home Team Science and Technology Agency

Incoming AI Engineer Intern

Aug 2026 - Dec 2026

- Selected for a role focused on multimodal AI prototyping including LLMs, vision-language models, transcription, and text-to-speech systems
- Role includes cloud-based ML development using AWS, Docker, SageMaker, and web application engineering

Singtel

Software Engineer Intern

May 2026 - Jul 2026

- Analyzed the Singtel e-commerce checkout sequence using Postman and browser network tools, inspecting API request/response flows, headers, and HTTP status codes within the sales channel
- Refactored a Spring Boot service to make the phone-number batch retrieval dynamically configurable via application.yaml, replacing hardcoded fetch-size values with runtime parameters and fallback defaults
- Gained hands-on exposure to CI/CD workflows using Bamboo and Bitbucket across non-production environments
- Utilized AWS CloudWatch Log Insights to monitor application logs and write queries for operational analysis
- Worked in an Agile team using Jira and Confluence, participating in sprint standups and story-point estimations

YouthTechSG

Open-Source Contributor

May 2026 - Oct 2026

- Contributed to FormSG, an open-source Singapore government platform, migrating the Payments landing page and Not Found page from hardcoded text to the i18n internationalization system
- Implemented TypeScript translation interfaces, replacing hardcoded strings with key-based lookups
- Structured locale blueprints and fallback translations to guarantee no missing or blank text
- Followed open-source contribution workflows, submitting documented pull requests for code review and merging

PROJECTS

Trackly | React, Node.js, Express, MongoDB, JWT, TailwindCSS

- Built a full-stack internship tracking app with 50+ sign-ups, allowing users to track internship pipelines with full CRUD, search, filtering, and pagination through a responsive dashboard
- Implemented stateless JWT authentication with password hashing and protected routes
- Designed RESTful APIs and MongoDB data models using a modular MVC structure
- Deployed frontend on Vercel and backend on Render, storing secrets in environment variables

Optima | React, TypeScript, Node.js, PostgreSQL, Prisma, Docker

- Built a full-stack platform ranking 133 Singapore secondary schools using a Rank Order Centroid (ROC) weighted scoring engine that converts user-ranked preferences into weights over commute, CCAs, programs, and subjects
- Integrated live OneMap public-transport routing, with 30-day database caching in PostgreSQL via Prisma ORM to eliminate redundant API calls
- Engineered a constraint-filtering pipeline with automated bottleneck detection that proposes quantified relaxations when no schools match a user's must-have criteria

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, Java, SQL, HTML/CSS

Frameworks: React, Node.js, Express, Spring Boot, Tailwind CSS

Tools/Platforms: AWS, PostgreSQL, MongoDB, Prisma, Docker, Git, Bitbucket, Bamboo, Postman, Jira, Confluence

Certifications: AWS Solutions Architect - Associate (Ongoing)

CO-CURRICULAR ACTIVITIES

Leadership: Hall Orientation Chief Group Leader, Hall Block Head, Safety Officer for Outdoor Adventure Club

Interests: Hiking, Scuba Diving, Soccer, Travelling, Running